



TECHNOLOGICAL UNIVERSITY
(MANDALAY)



DEPARTMENT OF CHEMICAL
ENGINEERING

EXTRACTION OF BIOACTIVE COMPOUNDS FROM DRIED TEA LEAVES

OBJECTIVES

- To extract soluble compounds from dried tea leaves
- To understand the principle and operation of Soxhlet extraction
- To study the extraction process using a suitable solvent
- To obtain a concentrated tea extract

INTRODUCTION

Tea leaves (*Camellia sinensis*) are rich in naturally occurring compounds such as polyphenols, catechins, flavonoids, caffeine, chlorophyll, and amino acids. Epigallocatechin gallate (EGCG) is one of the major catechins and is known for its strong antioxidant activity.

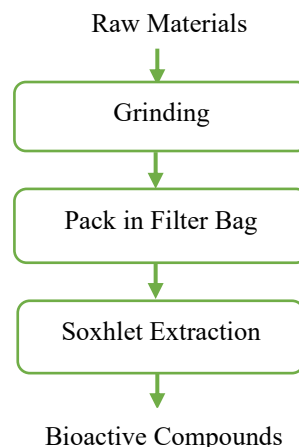
SAMPLE PREPARATION



SOXHLET EXTRACTION

Soxhlet extraction is a solid-liquid extraction technique used to continuously extract bioactive compounds from a solid material using suitable solvent.

FLOW CHART



Experimental Work



Possible Applications of the Extracted Materials

- Perfume
- Cosmetics
- Nutraceuticals (Omega3 fatty acids)
- Natural antioxidants (Carotenoids)
- Pharmaceutical products

CONCLUSION

The soluble bioactive compounds are extracted by Soxhlet extraction method which is efficient and convenient for dried tea leaves. It is carried out at temperature (78°C) for 3-4 hours.